Observational Study

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Assessing the readability, reliability, and quality of artificial intelligence chatbot responses to the 100 most searched queries about cardiopulmonary resuscitation

An observational study

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Abstract

This study aimed to evaluate the readability, reliability, and quality of responses by 4 selected artificial intelligence (AI)-based large language model (LLM) chatbots to questions related to cardiopulmonary resuscitation (CPR). This was a cross-sectional study. Responses to the 100 most frequently asked questions about CPR by 4 selected chatbots (ChatGPT-3.5 [Open AI], Google Bard [Google AI], Google Gemini [Google AI], and Perplexity [Perplexity AI]) were analyzed for readability, reliability, and quality. The chatbots were asked the following question: "What are the 100 most frequently asked questions about cardio pulmonary resuscitation?" in English. Each of the 100 queries derived from the responses was individually posed to the 4 chatbots. The 400 responses or patient education materials (PEM) from the chatbots were assessed for quality and reliability using the modified DISCERN Questionnaire, Journal of the American Medical Association and Global Quality Score. Readability assessment utilized 2 different calculators, which computed readability scores independently using metrics such as Flesch Reading Ease Score, Flesch-Kincaid Grade Level, Simple Measure of Gobbledygook, Gunning Fog Readability and Automated Readability Index. Analyzed 100 responses from each of the 4 chatbots. When the readability values of the median results obtained from Calculators 1 and 2 were compared with the 6th-grade reading level, there was a highly significant difference between the groups ($P < .001$). Compared to all formulas, the readability level of the responses was above 6[th] grade. It can be seen that the order of readability from easy to difficult is Bard, Perplexity, Gemini, and ChatGPT-3.5. The readability of the text content provided by all 4 chatbots was found to be above the 6th-grade level. We believe that enhancing the quality, reliability, and readability of PEMs will lead to easier understanding by readers and more accurate performance of CPR. So, patients who receive bystander CPR may experience an increased likelihood of survival.

Abbreviations: AI = artificial intelligence, CPR = cardiopulmonary resuscitation, LLM = large language model, PEMs = patient education materials.

Keywords: Artificial intelligence, Bard, chatbot, ChatGPT, Gemini, Perplexity, quality, readability, reliability

1. Introduction

Nowadays, the Internet is a preferred source of medical information owing to easy access to information, low cost, and accessibility.[1] In early 2023, there were 5.16 billion Internet users worldwide, equivalent to 64.4% of the world population at the time.[2] Studies reported that the Internet is the fastest growing source of health information and is used more by patients with chronic conditions, the uninsured, and those living far from their physicians.[1,3,4] Until recently, traditional search engines, such as Google and Yahoo, were used to access medical information. However, with the release of version 3.5 of Chat Generative Pre-Trained Transformer (ChatGPT), a large language model (LLM) powered by artificial intelligence (AI) in 2022, a new alternative

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source of information has emerged.[5] ChatGPT reportedly reached an average of 55 million daily visitors in June 2023, with an estimated 1.66 billion visitors in the same month.[6] ChatGPT is an AI chatbot created by OpenAI.[5] Chatbots are intelligent systems developed using rule-based or self-learning (AI) methods.[6] Chatbots receive user queries as input and respond via voice or messaging.[5] They are increasingly used by individuals and organizations due to their 24/7 availability, low cost, ease of use and ability to provide fast output.[5]

Perplexity AI (2022), and Google Bard (2023) are other products that entered the market as "chatbots" that use machine learning and natural language processing (NLP) to respond to user questions.[7] Perplexity is based on OpenAI GPT technology but is specially trained as a chatbot. It provides responses to queries and prompts and includes links to citations and related topics.[7,8] Google Bard, another AI-based search engine, was developed as a competitor to ChatGPT. Bard was trained with similar data and has the same goals as the ChatGPT. The main difference is that Bard is based on the language model for the dialogue application (LaMDA) family of LLMs.[8] Bard AI was rebranded as Google Gemini by Google AI (Mountain View, CA) in February 2024.[9] At the central of its design, the Gemini AI model is a "native multimodal" model that allows it to process and learn from various types of data, including text, audio, and video. Google Gemini is capable of analyzing complex data sets such as bottom charts and images, which is a significant improvement over previous Bard AI models.[9] There are very few publications in the literature investigating the potential benefits of Gemini for healthcare professionals in diagnosis and treatment.[9–11]

Although, these programs have gained popularity as a valuable tool among those who want to obtain information from Internet sources, caution is necessary when utilizing chatbots for medical information.[12] Also, there are potential risks in NLP associated with factual errors, often referred to as "hallucinations," that produce content that does not make sense or is not accurate in relation to certain sources.[5,13] Despite the potential positivity of chatbots in answering real questions, they have been reported to give inaccurate but fluent and confident answers.[5] There are concerns such as limitations in the accuracy and timeliness of information obtained from chatbots. It should be known that chatbots, in their current form, should not be the decision-making authority on healthcare and cannot replace consultations with medical professionals on issues such as evaluating treatment options and guiding the patient in decision-making.[5,12,13] Given the diversity of information sources and their widely varying quality, it is important to consider the ways in which individuals acquire health information.[1,12] It may be necessary to question the accuracy, reliability, and readability of information obtained through AI-based search engines, a new alternative information source.[12]

Out-of-hospital cardiac arrest (OHCA) is a major public health problem among adults, with an average global incidence of 55 OHCA cases per 100,000 person-years.[14] In Europe, the annual incidence of OHCA is 67 to 170 per 100,000 person-years. The rate of bystander cardiopulmonary resuscitation (CPR) was reported to be 58% (range 13%–83%), with variation between and within countries.[15] Conditions requiring CPR include a wide range of treatment options.[16–19] In the literature, Chat GPT and other AI-based LLM programs were asked questions about different medical topics and the accuracy and readability of the responses were investigated.[2,20–28] However, we did not find any study in the literature evaluating the accuracy and readability of CPR-related recommendations obtained from AI-based LLMs. We hypothesize that the answers of AI-based LLMs to the most frequently asked queries about CPR will be difficult to read and have moderate reliability and quality. To test this hypothesis, we evaluated the readability, reliability, and quality of responses provided to the most frequently asked queries about CPR by ChatGPT-3.5, Google Bard, Google Gemini, and Perplexity, which we selected as examples of AI-based LLMs for the purpose of this study.

2. Methods

2.1. Ethics

Since ChatGPT-3.5, Google Bard, Google Gemini, and Perplexity used in this study are publicly available applications and there were no human/animal participants, ethics committee approval and consent were not required.

2.2. Study setting

This study was conducted in September 2023. Gemini was later included in the study, as Bard AI was rebranded as Google Gemini by Google AI in February 2024.[9] Google Gemini queries were inputted in March 2024. This was an observational cross-sectional study.

2.3. Data sources and sampling

First of all, the aim was to determine which questions are most frequently asked in AI-based LLMs regarding CPR. For this purpose, "What are the 100 most frequently asked questions about cardio pulmonary resuscitation?" the question was asked separately in English to the 3 chatbots, available publicly for free [ChatGPT-3.5 (Open AI), Google Bard (Google AI) and Perplexity (Perplexity AI)] on September 10, 2023. A total of 100 queries were generated by evaluating the responses obtained in terms of similarity and repetition of queries. These 100 queries, written in English, were asked to individually each of the 3 chatbots. While this manuscript was written, the Bard AI was rebranded as Gemini by Google AI (Mountain View, CA).[9] The newly launched Gemini AI has been added to the study to keep up with version changes and updates to chatbots. Asking 100 questions about CPR to Gemini AI and readability analysis were conducted from March 27 to 30, 2024.

Unlike the GPT-3.5 version, which is publicly available for free, the advanced GPT-4 version is accessible through a paid service.[29,30] Firstly, those who wish to try the GPT language model must obtain a paid subscription for Chat-GPT+. ChatGPT Plus is available for $20 per month, and GPT-4 Team is available through a $25 per month subscription. There is no free access to GPT-4.[31] The fact that ChatGPT-4 requires a paid subscription is a factor that makes it difficult for people with a low socio-economic status to use GPT-4. Although GPT-4 is an up-to-date version, it was excluded from the scope of this study as it is less accessible due to paid subscription. This study only includes AI-based LLMs that are freely available. All responses were recorded and evaluated. The number of questions in this study was determined as in the study by Holmes et al.[28] The 100 queries and 400 responses are available from the web archive located at: https://archive.org/details/assessing-of-the-ai-chatbots-responses-to-the-100-most-searched-queries-about-ca/page/n11/mode/2up

2.4. Reliability and quality assessment

During text analysis, the responses obtained in the first stage were evaluated in terms of reliability and quality. The modified DISCERN Questionnaire, Journal of the American Medical Association (JAMA), and Global Quality Score (GQS) were used for each question. For the Modified DISCERN Scale, the following questions were asked about each response: "does the text address areas of controversy/uncertainty?," "are additional sources of information listed for patient reference?," "is the information provided balanced and unbiased?," "are valid sources cited (valid studies, doctors)?," and "is the text clear and understandable?" Each question was evaluated based on one point. According to this evaluation criterion, the text received a minimum score of 0 and a maximum score of 5.[32] For the Journal of American Medical Association (JAMA) quality test criteria, the following points were evaluated. Authorship (1 point): authors and contributors, their links, and relevant credentials must be provided; citation (1 point): references and sources must be listed for all content; disclosure (1 point): conflicts of interest, funding, sponsorship, advertising, endorsement, and text ownership must be fully disclosed; and timeliness (1 point): dates when content was published and updated should be indicated.[33] The Global Quality Score (GQS) system is as follows: 5 points: excellent quality and excellent flow, very beneficial for patients; 4 points: good quality and overall good flow, beneficial for patients; 3 points: moderate quality, suboptimal flow, slightly beneficial for patients; 2 points: overall poor quality and poor flow for very limited use for patients; and 1 point: poor quality, poor flow of the site, not at all useful for patients."[32]

2.5. Readability assessment

In the second stage of text analysis, the text was evaluated for readability. Each response received from the AI chatbots was evaluated for readability using 2 different readability-calculation websites. First, the 4 different responses given by the 4 chatbots to each query were entered into http://readabilityformulas.com/ (Calculator 1). The Flesch Reading Ease Score (FRES), Flesch-Kincaid Grade Level (FKGL), Simple Measure of Gobbledygook (SMOG), Gunning Fog Readability (GFOG), Coleman-Liau Readability Index (CLI), Linsear Write (LW) readability formula, and Automated Readability Index (ARI) results were calculated according to the data. Secondly, similar to Calculator 1, queries were entered into Calculator 2 and recalculated. The readability of each response to the queries was calculated using 6 validated objective indices (FRES, FKGL, GFOG index, SMOG Index, CLI, and ARI) using https://www.online-utility.org/english/readability_test_and_improve.jsp (Calculator 2) for a second readability assessment of the responses received from the AI chatbots. The flow chart of question selection and evaluation of responses is shown in Figure 1. These algorithms have been validated in the literature and are used to assess the readability of patient-centered medical texts in healthcare.[21–27,34] The algorithms and descriptions of each readability scale are listed in Table 1. The readability of the responses was recorded by 5 independent senior anesthesia residents (İ.E., F.K., N.S., M.O., and E.İ.). These data were evaluated by 2 senior lecturers in anesthesiology (D.Ö.A and V.H.). In addition, the reliability and quality of the responses were recorded and evaluated separately by D.Ö.A and V.H. It was calculated by taking the mean of the results obtained from both. The results are shown in Tables 2 to 5.

2.6. Readability tests

The Flesch-Kincaid readability tests were divided into 2 groups. FRES indicates the degree of readability of the text, whereas FKGL is related to the level of education of the person.[35,36] *FRES formula;* This test score is based on a ranking scale from 0 to 100 and the higher the score, the easier it is to understand. Readability Score 70 to 79: fairly easy, 60 to 69: standard (plain English), 50 to 59: fairly difficult, and 30 to 49: difficult. A score of 90 to 100 is considered comprehensible by those with an average of 5 years of education, 70 to 80 by those with an average of 7 years of education, 60 to 70 by those with an average of 8 to 9 years of education, 50 to 60 by those with an average of 10 to 12 years of education, and 30 to 50 by those with an average of 13 to 16 years of education (college).[35,36,39] *FKGL formula;* indicates the academic grade level required to understand the written material. For example, an FKGL score of 6.4 indicates that the material is written at a 6th to 7th-grade reading level.[35–39] *SMOG index*: This estimates the number of years of education a person needs to understand a text. It scores a text according to the complexity of its sentences and words. The higher the number of polysyllabic words, the higher the score.[35–39] *Gunning Fog Test:* Sentence length, and percentage of polysyllabic words are important for this test. The ideal score for the Fog Index is 7 or 8. Above 12, it is difficult for most people to read.[35–39] *Coleman-Liau Index (CLI):* It was designed to assess the grade level in US schools required to understand the text. The score received indicates the US school grade level a person needs to understand the text. Therefore, if the score is 9.5, text should be easily understood by students in grades 9 or 10.[37–39] *The Automated Readability Index (ARI):* This assesses the US school grade level required to read a text. It

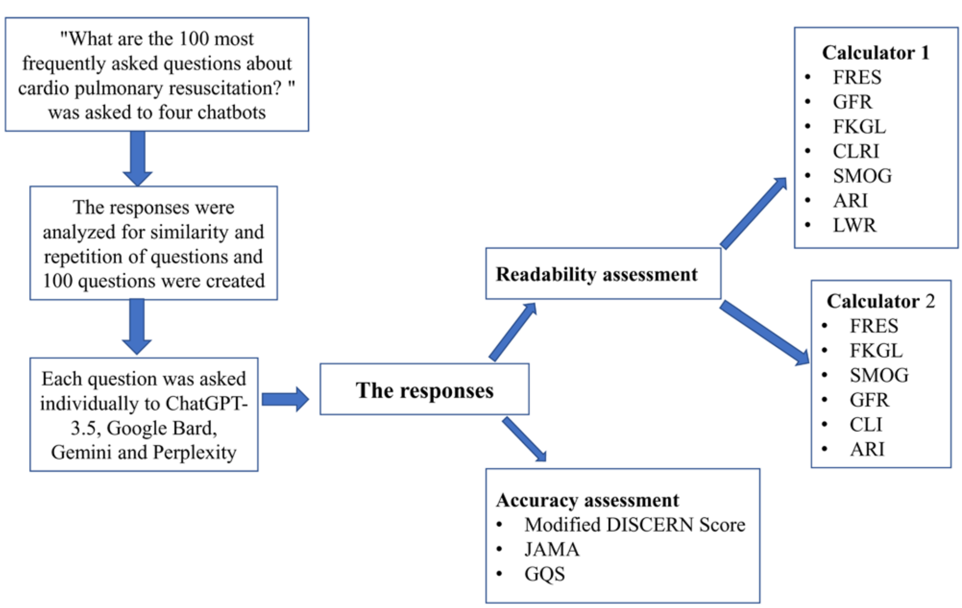

Figure 1. Flow chart for question selection and evaluation of responses.

Table 1

Readability tools, formulas and features.

Index	Formula	Historical use	Features
FRES[35–39]	$206.835 - 1.015\left(\frac{\text{words}}{\text{sentences}}\right) - 84.6\left(\frac{\text{syllabes}}{\text{words}}\right)$	Evaluates the readability of various written materials (newspaper articles etc). Developed in the late 1940s,	Scores range from 0 to 100. The higher score, the easier it is to understand. Low scores indicate that the text is complex to understand
FKGL[35–39]	$0.39\left(\frac{\text{words}}{\text{sentences}}\right) + 11.8\left(\frac{\text{syllables}}{\text{words}}\right) - 15.59$	Developed under contract to the US Navy in 1975	Indicates the academic grade level required to understand written material
GFOG[35–39]	$0.4\left[\left(\frac{\text{word}}{\text{sentences}}\right) + 100\left(\frac{\text{Complex words}}{\text{words}}\right)\right]$	Applicable across various disciplines, originally intended for public newspapers, developed in the early1950	It estimates the number of years of education a person needs to understand a text
SMOG[35–39]	$1.0430\sqrt{S\left(\frac{30}{\text{sentences}}\right)} + 3.1291$ *S = Number of words with ≥ 3 syllables*	Assesses text readability by counting polysyllabic words. Developed in the late 1960	It measures how many years of education the average person needs to have to understand a text
CLI[35, 37, 3954]	(0.0588 × L)–(0.296 × S)–15.8 *L* = avarage number of letters per 100 words. *S* = avarage number of sentences per 100 words	Developed for United States Office of Education Textbooks in 1975	Measures the grade level needed to comprehend a text and provides a corresponding US school level
ARI[35,38,39]	$4.71\left(\frac{\text{characters}}{\text{words}}\right) + 0.5\left(\frac{\text{words}}{\text{sentences}}\right) - 21.43$	Created for and validated by the US military in 1967, it was designed to assess the grade level required to comprehend their technical publications	Evaluates the grade level in US schools required to be able to read a text. The more characters there are, the more difficult the word is
LW[38,39]	$\frac{E+3C}{\text{Number of sentences}}$ Result • If > 20, divide by 2 • If ≤ 20, subtract 2, and then divide by 2	Developed in the late 1970 to calculate the readability of texts, especially technical manuals, for the United States Air Force	Provides a rough estimate of the grade level required to understand the text

ARI = automated readability index, C = number of complex words with ≥3 syllables, CLI = Coleman-Liau Index, E = Number of easy words with 1 or 2 syllables, FKGL = Flesch-Kincaid Grade Level, FRES = Flesch Reading Ease Score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook.

differs from other methods in that it counts characters rather than syllables. The more characters there are, the harder the word is to understand.[37–39] The scale was designed to align with the US education system, where grades 1 through 14 range from the first year of elementary school to the last year of college. *Linsear Write Formula (LW):* This provides a rough estimate of the grade level required to understand the text. For example, using the LW formula, a score of 8 indicates that a student in the eighth grade (usually 13–14 years old) should be able to understand the text. Similarly, a score of 12 indicates that the text is suitable for senior high school students.[38,39]

2.7. Statistical analysis

SPSS Windows 24.0 (SPSS Inc., Chicago, IL) was used for statistical analysis. Frequency data are expressed as numbers (n) and percentages (%). Continuous data are presented as median (minimum-maximum). Chi-square test and Fisher exact test were used to compare frequency variables. Mann-Whitney U and Wilcoxon tests were used to compare continuous data. To determine the consistency of the calculators, intraclass correlation coefficient (ICC) analysis was performed for each formula. A *P* value of <.05 was determined as a significant difference.

3. Results

The readability scores for the responses of ChatGPT-3.5, Google Bard, Google Gemini, and Perplexity programs to the queries included in the 100 most asked questions about CPR were determined using Calculators 1 and 2. The readability scores for the text were statistically compared with the 6th-grade reading level and analyzed. Tables 2, 3 and 4 present the results.

3.1. ChatGPT-3.5 analysis results using average results from Calculator 1 and Calculator 2

In the text readability analysis of 100 ChatGPT-3.5 outputs obtained in response to 100 queries, the median FRES was 42.09 (6.30–64.19) and the median GFOG was 14.33 (9.71–20.53). In addition, the median FKGL, SMOG, CLI, and ARI means were 11.70 (7.64–18.22), 12.10 (8.92–17.74), 12.50 (8.54–18.48), and 11.66 (6.59–18.66) years of education, respectively (Table 4).

3.2. Bard analysis results using average results from Calculator 1 and Calculator 2

In the text readability analysis of Bard 100 responses, the median FRES was 67.85 (27.46–84.79) and GFOG median was 10.14 (5.59–20.98). In addition, the median FKGL and SMOG were 7.73 (3.41–18.30) and 8.84 (5.57–15.87) years of education, respectively. CLI was 8.25 (4.87–14.38) years of education and ARI was 7.17 (2.64–20.71) years of education (Table 4).

3.3. Gemini analysis results using average results from Calculator 1 and Calculator 2

In the text readability analysis of Gemini 100 responses, the median FRES was 45.30 (6.32–78.55) and GFOG median was 13.45 (8.65–18.36). In addition, the median FKGL and SMOG were 10.54 (6.13–15.02) and 10.95 (6.00–14.41) years of education, respectively. CLI was 13.31 (8.98–20.32) years of education and ARI was 11.18 (7.41–16.20) years of education (Table 4)

3.4. Perplexity analysis results using average results from Calculator 1 and Calculator 2

In the text readability analysis of 100 Perplexity responses, the median FRES was 53.98 (6.66–82.36) and the median GFOG was 12.58 (5.67–22.21). In addition, the median FKGL and SMOG were 10.12 (4.54–18.54) and 10.61 (5.24–18.84) years of education, respectively. CLI was 10.90 (6.54–21.39) years of education and ARI was 10.33 (3.76–19.50) years of education (Table 4).

Table 2

Readability indices for ChatGPT 3.5, Bard, Gemini, and Perplexity responses to the 100 most asked questions about cardiopulmonary resuscitation, and statistical comparison of text content to 6th-grade reading level [median (minimum–maximum)] by using calculator 1 (https://readabilityformulas.com/free-readability-formula-tests.php).

CHAT GPT Calculator 1 statistics	ChatGPT 3.5	Google Bard	Google Gemini	Perplexity	Chat GPT C6thGRL (*P*)*†	Bard C6thGRL (*P*)*†	Gemini C6thGRL (*P*)*†	Perplexity C6thGRL (*P*)*†	Between ChatGPT and Bard††	Between ChatGPT and Gemini††	Between Bard and Perplexity††	Between Bard and Gemini††	Between ChatGPT and Perplexity††	Between Gemini and Perplexity††
FRES	43.95	71,00	46.00	53.40	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.001**	**.031**
	(8.90–66.30)	(29.70–88.00)	(3.00–79.00)	(−12.0 to 86.10)			**<.001**			.289				
GFOG	14.40	10.15	14.0	13.00	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.012**	.225
	(9.20–20.80)	(6.00–21.80)	(8.90–19.40)	(5.60–22.20)			**<.001**			.152				
FKGL	11.30	7.00	10.54	10.20	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.007**	.892
	(7.40–17.80)	(2.88–18.00)	(5.82–15.49)	(4.00–17.95)			**<.001**			**.001**				
CLI	12.00	8.00	13.32	11.00	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.003**	**<.001**
	(8.00–18.57)	(5.00–14.00)	(8.35–22.35)	(6.00–25.14)			**<.001**			**.012**				
SMOG	10.75	7.24	9.97	9.43	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**<.001**	.054
	(7.40–15.60)	(3.90–14.60)	(6.00–14.06)	(3.90–17.10)			**<.001**			**.001**				
ARI	11.70	7.05	11.83	10.50	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.016**	**.011**
	(6.30–18.60)	(2.40–20.70)	(7.78–18.18)	(3.70–21.58)			**<.001**			.962				
LW	12.10	7.70	10.13	10.80	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**<.001**	.207
	(6.59–19.40)	(2.96–25.30)	(5.59–15.93)	(4.00–22.90)			**<.001**			**<.001**				
Grade level	12.00	8.00	11.00	11.00	**<.001**	**<.001**		**<.001**	**<.001**		**<.001**	**<.001**	**.007**	.062
	(7.00–18.00)	(4.00–18.00)	(8.00–16.00)	(5.00–19.00)			**<.001**			.244				
Reading level	n (%)	n (%)	n (%)	n (%)										
Easy to read	0 (0)	5 (5.0)	0 (0)	1 (1.0)					**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**
Fairly easy to read	0 (0)	40 (40.0)	0 (0)	5 (5.0)										
Standard/Average	10 (10.0)	28 (28.0)	4 (4.1)	24 (24.0)										
Fairly difficult to read	28 (28.0)	22 (22.0)	45 (45.5)	27 (27.0)										
Difficult to read	44 (44.0)	5 (5.0)	22 (22.2)	20 (20.0)										
Very difficult to read	18 (18.0)	0 (0)	19 (19.2)	23 (23.0)										
Impossible to comprehend	0 (0)	0 (0)	9 (9.1)	0 (0)										
Readers age	n (%)	n (%)	n (%)	n (%)										
8–9 yrs. old	0 (0)	6 (6.0)	0 (0)	1 (1.0)					**<.001**	**.019**	**<.001**	**<.001**	.255	**<.001**
(fourth and fifth graders) 10–11 yrs. olds	0 (0)	17 (17.0)	0 (0)	4 (4.0)										
(fifth and sixth graders) 11–13 yr old	1 (1.0)	21 (21.0)	0 (0)	4 (4.0)										
(sixth and seventh graders) 12–14 yr old	7 (7.0)	16 (16.0)	0 (0)	11 (11.0)										
(seventh and eighth graders) 13–15 yr old	4 (4.0)	15 (15.0)	5 (5.1)	9 (9.0)										
(eighth and ninth graders) 14–15 yr old	12 (12.0)	13 (13.0)	13 (13.1)	11 (11.0)										
(ninth to tenth graders) 15–17 yr old	22 (22.0)	8 (8.0)	33 (33.3)	20 (20.0)										
(tenth to eleventh graders) n (%) 17–18 yr old	18 (18.0)	2 (2.0)	20 (20.2)	11 (11.0)										
(twelfth graders) 18–19 yr old	14 (14.0)	0 (0)	12 (12.1)	10 (10.0)										
(college level entry) 21–22 yr old	7 (7.0)	0 (0)	12 (12.1)	5 (5.0)										
(college level) College graduate	15 (15.0)	2 (2.0)	4 (4.0)	14 (14.0)										

P values in bold are statistically significant.

ARI = automated readability index, CLI = Coleman–Liau index, FKGL = Flesch–Kincaid Grade Level, FRES = Flesch Reading Ease Score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook.

*C6thGRL(*P*): Comparison of the responses according to 6th grade reading level (*P*).

†Wilcoxon test.

††Chi-square test for categorical variables and Mann–Whitney *U* test for continuous variables.

3.5. Comparison of the readability of all 4 groups using average results from Calculator 1 and Calculator 2

When comparing the readability of responses across all 4 groups using average results from Calculator 1 and Calculator 2, significant differences were observed between certain groups. The significant differences were found between: ChatGPT-3.5 and Bard groups ($P < .001$), ChatGPT-3.5 and Perplexity groups ($P < .001$), Bard and Gemini groups ($P < .001$), Bard and Perplexity groups ($P < .001$) (Table 4). Furthermore, when ChatGPT and Gemini groups were compared, there were no significant differences for CLI and ARI ($P > .05$). In addition, there were no significant differences for GFOG, FKGL, and SMOG when comparing the Perplexity and Gemini groups ($P > .05$) (Table 4).

According to the readability scores obtained, all readability formulas except CLI are ranked in order of readability from easy to difficult: Google Bard, Perplexity, Google Gemini, and ChatGPT-3.5. However, according to the CLI readability formula, the ranking differs slightly: Google Bard, Perplexity, ChatGPT-3.5, and Google Gemini (Table 4).

3.6. Comparison of ChatGPT, Bard, Gemini, and Perplexity responses according to the recommended 6th-grade reading level

When the median readability indices of all responses were compared with the sixth-grade reading level, a statistically significant difference was observed for all formulas compared to the sixth-grade level ($P < .001$). When compared with all formulas, the readability level of the responses was above the sixth-grade level. Statistically significant results were obtained when Calculator 1, Calculator 2, and the average of both calculators were compared ($P < .001$) (Tables 2–4).

3.7. Reliability and quality of responses

Bard provided a source for 1 question. Gemini provided links to some questions as an alternative source of information. Gemini did not answer one question at all (question 8: How do I perform CPR on a person with a foreign body airway obstruction?). In response to this question, Gemini reported that it was not programmed to assist with that. Also in some question, it stated that it was an informative LLM, could not provide medical advice and the information provided was for educational purposes only. In Perplexity, there were usually 5 or 6 sources for each response. The sources varied from videos, illustrated narratives, scientific texts (YouTube videos, public information pages of associations, scientific articles on PubMed, etc). Videos and images related to CPR were presented. For modified DISCERN, there was a statistical significance for all groups ($P < .05$) (Table 5).

3.8. Reliability and quality of responses

In the outputs generated by ChatGPT-3.5, no sources were provided. Google Bard offered a source for only one question. Gemini provided some links as an alternative source of information in response to some questions. However, Gemini did not answer one question at all (question 8: How do I perform CPR on a person with a foreign body airway obstruction?), stating that it was not programmed to assist with that.

Additionally, in some of the responses, Gemini provided a disclaimer stating that it was an informative LLM and could not offer medical advice; the information provided was solely for educational purposes. In contrast, Perplexity typically included 5 or 6 sources for each response, which ranged from videos and illustrated narratives to scientific texts (such as YouTube videos, public information pages of associations, and scientific articles on PubMed). Videos and images related to CPR were also presented.

For the modified DISCERN, there was a statistical significance observed for all groups ($P < .05$) (Table 5).

3.9. Intraclass correlation coefficients

FRES, GFOG, FKGL, CL, SMOG, and ARI were calculated using 2 different calculators (https://readabilityformulas.com/free-readability-formula-tests.php, https://www.online-utility.org/english/readability_test_and_improve.jsp).

3.9.1. ICCE for chat GPT. The intraclass correlation coefficients were 0.991 for FRES, 0.977 for GFOG, 0.993 for KFGL, 0.948 for CL, 0.977 for SMOG, and 0.985 for ARI.

3.9.2. ICCE for Bard. The intraclass correlation coefficients were 0.984 for FRES, 0.977 for GFOG, 0.988 for KFGL, 0.977 for CL, 0.972 for SMOG, and 0.990 for ARI.

3.9.3. ICCE for Gemini. The intraclass correlation coefficients were 0.947 for FRES, 0.857 for GFOG, 0.877 for KFGL, 0.930 for CL, 0.710 for SMOG, and 0.840 for ARI.

3.9.4. ICCE for Perplexity. The intraclass correlation coefficients were 0.918 for FRES, 0.920 for GFOG, 0.913 for KFGL, 0.912 for CL, 0.954 for SMOG, and 0.853 for ARI.

3.9.5. ICCE for GQS, JAMA, and mDISCERN. The intraclass correlation coefficients were 0.942 for GQS, 0.965 for JAMA, and 0.914 for mDISCERN.

4. Discussion

In this study, the outputs of all 4 AI-based LLMs responding to the query "What are the 100 most asked questions about cardiopulmonary resuscitation?" were evaluated for readability, quality, and reliability. As far as we know, this is the first study to compare responses generated by ChatGPT-3.5, Google Bard, Google Gemini, and Perplexity with CPR-related queries. It is also, to our knowledge, the first study to evaluate the readability, reliability and quality of GeminiAI. The patient education materials (PEM) obtained in our study were mostly favorable in terms of quality and reliability. However, when the PEM obtained from all 4 chatbots was analyzed in terms of readability, they were found to be above the 6th-grade level.

4.1. Readability assessment

Readability refers to how easy or difficult texts are understood by the reader. Research shows that readability of a given text is closely related to its comprehension.[40] This is particularly important in health-related texts, as they have the potential to harm the patient if not understood or misunderstood.[41] CPR is a set of interventions used to restore oxygen and circulation to the body during cardiac arrest. Emergency CPR can double or triple the chance of survival after cardiac arrest.[42] The importance of the reader being able to read, understand, and apply CPR information correctly is clear. There are publications in the literature reporting that AI-based LLM outputs are difficult or very difficult to read, and that good training is required to understand the material.[2,20,24] Momenaei et al evaluated the relevance and readability of medical information provided by ChatGPT-4 regarding common vitreoretinal surgeries and reported that according to the mean scores for FKGL and FRES, the responses were difficult or very difficult for ordinary person to read and that a university degree was required to understand the material.[20] In a similar study, Robinson et al compared the accuracy and readability of responses to common

Table 3

Readability indices for ChatGPT-3.5, Bard, Gemini, and Perplexity responses to the 100 most asked questions about cardiopulmonary resuscitation, and statistical comparison of text content to 6th-grade reading level [median (minimum-maximum)] by using Calculator 2. (https://www.online-utility.org/english/readability_test_and_improve.jsp).

Calculator 2 statistics	ChatGPT	Bard	Gemini	Perplexity	ChatGPT C6thGRL (*P*)*†	Bard C6thGRL (*P*)*†	Gemini C6thGRL (*P*)*†	Perplexity C6thGRL (*P*)*†	Between ChatGPT and Bard (*P*)††	Between ChatGPT and Gemini (*P*)††	Between ChatGPT and Perplexity (*P*)††	Between Bard and Gemini (*P*)††	Between Bard and Perplexity (*P*)††	Between Perplexity and Gemini (*P*)††
FRES	40.20 (3.40–62.08)	64.94 (25.22–81.58)	45.60 (9.64–78.10)	51.97 (7.78–78.62)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**.002**	**.001**	**<.001**	**<.001**	**.013**
GFOG	14.15 (10.16–20.25)	9.995 (5.17–20.15)	12.87 (8.00–17.71)	12.23 (5.74–22.22)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**.012**	**<.001**	**<.001**	.059
FKGL	12.02 (7.88–18.64)	8.07 (3.94–18.59)	10.49 (6.00–15.21)	10.05 (5.08–19.17)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**.007**	**<.001**	**<.001**	.121
CLI	12.91 (9.04–18.96)	8.52 (4.73–14,75)	13.23 (9.14–18.41)	10.72 (6.08–18.14)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	.453	**<.001**	**<.001**	**<.001**	**<.001**
SMOG	13.39 (10.44–19.97)	10.26 (6.95–17.14)	12.08 (6.00–16.02)	11.86 (6.57–20.57)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	.723
ARI	11.69 (6.22–18.71)	7.18 (2.51–20.72)	10.58 (6.20–15.91)	9.57 (3.82–19.59)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**.016**	**<.001**	**<.001**	**.021**

P values in bold are statistically significant.
ARI = Automated Readability Index, CLI = Coleman–Liau Index, FKGL = Flesch–Kincaid Grade Level, FRES = Flesch Reading Ease Score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook.
*C6thGRL(*P*)**:** Comparison of the responses according to 6th grade reading level (*P*).
†Wilcoxon test.
††Chi-Square test for categorical variables and Mann–Whitney *U* test for continuous variables.

Table 4

Readability indices for ChatGPT 3.5, Bard, Gemini, and Perplexity responses to the 100 most asked questions about cardiopulmonary resuscitation and statistical comparison of text content to 6th-grade reading level [median (minimum–maximum)] using the average results obtained from Calculator 1 and Calculator 2.

Readability indexes	ChatGPT*	ChatGPT C6thGRL (*P*)**†	Bard*	Bard C6thGRL (*P*)**†	Gemini*	Gemini C6thGRL (*P*)**†	Perplexity*	Perplexity C6thGRL (*P*)**†	Between ChatGPT and Bard (*P*) ††	Between ChatGPT and Gemini ††	Between ChatGPT and Perplexity (*P*)††	Between Bard and Gemini (*P*)††	Between Bard and Perplexity (*P*)††	Between Perplexity and Gemini (*P*)††
FRES	42.09 (6.30–64.19)	**<.001**	67.85 (27.46–84.79)	**<.001**	45.30 (6.32–78.55)	**<.001**	53.98 (6.66–82.36)	**<.001**	**<.001**	**.046**	**<.001**	**<.001**	**<.001**	**.025**
GFOG	14.33 (9.71–20.53)	**<.001**	10.14 (5.59–21.98)	**<.001**	13.45 (8.65–18.36)	**<.001**	12.58 (5.67–22.21)	**<.001**	**<.001**	**.003**	**<.001**	**<.001**	**<.001**	.101
FKGL	11.70 (7.64–18.22)	**<.001**	7.73 (3.41–18.30)	**<.001**	10.54 (6.13–15.02)	**<.001**	10.12 (4.54–18.54)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	.444
SMOG	12.10 (8.92–17.74)	**<.001**	8.84 (5.657–15.87)	**<.001**	10.95 (6.00–14.41)	**<.001**	10.61 (5.24–18.84)	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	**<.001**	.212
CLI	12.50 (8.54–18.48)	**<.001**	8.25 (4.87–14.38)	**<.001**	13.31 (8.98–20.32)	**<.001**	10.90 (6.54–21.39)	**<.001**	**<.001**	.088	**<.001**	**<.001**	**<.001**	**<.001**
ARI	11.66 (6.59–18.66)	**<.001**	7.17 (2.64–20.71)	**<.001**	11.18 (7.41–16.20)	**<.001**	10.33 (3.76–19.50)	**<.001**	**<.001**	.097	**<.001**	**<.001**	**<.001**	**.017**

Calculator 1: https://readabilityformulas.com/free-readability-formula-tests.php
Calculator 2: https://www.online-utility.org/english/readability_test_and_improve.jsp
P values in bold are statistically significant.
ARI = Automated Readability Index, CLI = Coleman-Liau Index, FKGL = Flesch-Kincaid Grade Level, FRES = Flesch reading ease score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook.
*[(Calculator 1) + (Calculator 2)]/2.
C6thGRL(p):** Comparison of the responses according to 6th grade reading level **(p).**
†Wilcoxon test.
††Chi-Square test for categorical variables and Mann-Whitney U test for continuous variables.

Table 5

Comparison of modified DISCERN, JAMA, and Global Quality Scale (GQS) scores of ChatGPT-3.5, Bard, Gemini, and Perplexity responses.

	ChatGPT versus Bard			ChatGPT versus Perplexity			Bard vs Perplexity			ChatGPT vs Gemini			Bard vs Gemini			Perplexity vs Gemini		
	ChatGPT	Bard	P	ChatGPT	Perplexity	P	Bard	Perplexity	P	ChatGPT	Gemini	P	Bard	Gemini	P	Perplexity	Gemini	P
GQS, n (%)			**.008***			.055*			.185*									
1-point	0 (0)	1 (1)		0 (0)	0 (0)		1 (1)	0 (0)		0 (0)	0 (0)	.109*	1 (1)	0 (0)	**<.001***	0 (0)	0 (0)	**.046***
2-point	13 (13)	33 (33)		13 (13)	24 (24)		33 (33)	24 (24)		13 (13)	11 (11.1)		33 (33)	11 (11.1)		24 (24)	11 (11.1)	
3-point	81 (81)	61 (61)		81 (81)	66 (66)		61 (61)	66 (66)		81 (81)	73 (73.7)		61 (61)	73 (73.7)		66 (66)	73 (73.7)	
4-point	6 (6)	4 (4)		6 (6)	10 (10)		4 (4)	10 (10)		6 (6)	15 (15.2)		4 (4)	15 (15.2)		10 (10)	15 (15.2)	
5-point	0 (0)	1 (1)		0 (0)	0 (0)		1 (1)	0 (0)		0 (0)	0 (0)		1 (1)	0 (0)		0 (0)	0 (0)	
JAMA, n (%)			>.999**															
1-point	100 (100)	99 (99)		100 (100)	2 (2)	**<.001***	99 (99)	2 (2)	**<.001***	100 (100)	46 (46.5)	**<.00*1**	99 (99)	46 (46.5)	**<.001***	2 (2)	46 (46.5)	**<.001***
2-point	0 (0)	1 (1)		0 (0)	41 (41)		1 (1)	41 (41)		0 (0)	52 (52.5)		1 (1)	52 (52.5)		41 (41)	52 (52.5)	
3-point	0 (0)	0 (0)		0 (0)	33 (33)		0 (0)	33 (33)		0 (0)	1 (1)		0 (0)	1 (1)		33 (33)	1 (1)	
4-point	0 (0)	0 (0)		0 (0)	24 (24)		0 (0)	24 (24)		0 (0)	0 (0)		0 (0)	0 (0)		24 (24)	0 (0)	
m DISCERN, n (%)			**.018***			**< 001***			**<.001***			**.012***			**<.001***			**<.045***
1-point	1 (1)	1 (1)		1 (1)	1 (1)		1 (1)	1 (1)		1 (1)	0 (0)		1 (1)	0 (0)		1 (1)	0 (0)	
2-point	40 (40)	60 (60)		40 (40)	23 (23)		60 (60)	23 (23)		40 (40)	32 (32.3)		60 (60)	32 (32.3)		23 (23)	32 (32.3)	
3-point	59 (59)	39 (39)		59 (59)	54 (54)		39 (39)	54 (54)		59 (59)	58 (58.6)		39 (39)	58 (58.6)		54 (54)	58 (58.6)	
4-point	0 (0)	0 (0)		0 (0)	22 (22)		0 (0)	22 (22)		0 (0)	9 (9.1)		0 (0)	9 (9.1)		22 (22)	9 (9.1)	
5-point	0 (0)	0 (0)		0 (0)	0 (0)		0 (0)	0 (0)		0 (0)	0 (0)		0 (0)	0 (0)		0 (0)	0 (0)	

P values in bold are statistically significant.
*Chi-Square test.
**Fisher exact test.

Mohs surgery questions generated by ChatGPT and Google Bard to assess their suitability for patient education. In their study, they found that the mean FKDL ratings for the responses were higher than those of ChatGPT (10.3 ± 2.8, $P < .01$) and Google Bard (9.3 ± 1.3, $P < .01$). They reported that the PEM were too advanced for general patient education about medical information.[24] In their study evaluating the relevance, readability and other qualities of the urologic information generated by the ChatGPT, Davis et al reported that the mean (SD) FRES score of the outputs was 35.5 (10.2) and the mean FKGL score was 13.5 (1.74), indicating university-level readability of the responses, well above the recommended value.[2] In our study, the readability of all 4 AI-based LLM outputs, responding to the 100 most frequently asked queries about CPR, was calculated with 2 different calculators, and the data were evaluated using FRES, FKGL, SMOG, GFOG, CLI, LWR, and ARI. Compared to all formulas, the readability level of the responses was above 6th grade (Table 4). These scores obtained in our study, similar to the studies mentioned above, indicate that the responses are difficult to read.

It should be recognized that the value of knowledge is limited by the patient's ability to understand it. The American Medical Association (AMA) has recommended that all PEM be provided at 6th-grade reading level; however, the results of many studies about AI do not seem to follow this.[2,25,26] Pan et al analyzed the responses of 4 AI chatbots (ChatGPT-3.5, Perplexity, Chatsonic, and Bing AI) to the 5 most common cancer-related search queries and reported that the responses were of high quality but were written at college level according to the FKGL score. In their study, they reported that they should be used as an additional source rather than a primary source of medical information.[25] Similarly, in a study evaluating the readability of answers by entering queries about oropharyngeal cancer into ChatGPT-3.5, the average FRES and FKRGL scores were higher than the 11th-grade readability level and higher than the 6th-grade level recommended for patients.[23] The authors reported that ChatGPT responses may not educate patients to an appropriate degree, may misinform them completely, and that patients should be cautious when consuming chatbot-generated medical information because of a more difficult reading grade level than recommended for patient material.[23] In their study investigating the effectiveness of chatbots (ChatGPT, Bard, and Bing AI) for cataract patient education compared to traditional sources such as the American Academy of Ophthalmology (AAO) website, Yilmaz et al found the most accurate and comprehensive responses to cataract-related queries were provided by ChatGPT, Bard, Bing AI, and AAO website, respectively.[27] The most understandable and actionable responses were given by the AAO website, Bard, ChatGPT, and Bing AI, and the most readable responses were given by Bard, AAO website, ChatGPT, and Bing AI.[27] In our study, similarly, Bard 67.85 (27.46–84.79) received the highest score in the FRE score analysis, with the score for Perplexity 53.98 (6.66–82.36), for Gemini 45.3 (6.3–78.6) and for ChatGPT 42.09 (6.3–64.19) ($P < .001$). Accordingly, Bard responses had a standard reading level and required 8th to 9th grade level education to understand; Perplexity responses had fairly difficult reading level and required 10th to 12th grade level education to understand; Gemini and ChatGPT had a difficult reading level and required 13th to 16th grade (college) level education to understand. The United States Department of Health and Human Services recommends that PEM be written at the 6th to 7th-grade reading levels to make them accessible.[2] The easy readability of PEM will ensure that the reader is aware of new technologies and developments in CPR; therefore, the reader will understand how to perform CPR, thus increasing the survival rate of patients who receive bystander CPR.

Probably due to the recent launch of Google Gemini, we have not come across any study in the literature investigating the readability, reliability and quality of Gemini. In a study investigating the ability of ChatGPT-3.5, ChatGPT-4, and Google Gemini to analyze of retinal detachment cases and recommend the best possible surgical planning, it was reported that Gemini failed to respond in 5 cases.[11] In another study investigating the ability of ChatGPT-4 and Google Gemini to analyze glaucoma case descriptions and recommend an accurate surgical plan, the authors reported that Gemini was unable to complete the task in 16 cases (27%).[10] Similarly, in our study, unlike other chat robots, Gemini did not answer question number 8, even though it was asked several times. In response to this question, Gemini reported that it was not programmed to assist with that.

4.2. Reliability and quality of responses

The health-related information provided by AI-powered LLMs should offer high reliability and quality to the reader. In a recent study on the topic, researchers evaluated the responses of 4 AI chatbots (ChatGPT, Perplexity, Chat Sonic, and Microsoft Bing AI) to the 5 most common search queries from Google Trends related to prostate, bladder, kidney, and testicular cancers. AI chatbot responses had moderate to high information quality (median DISCERN score of 4 out of 5, range 2–5) and did not contain false information.[23] Similarly, Pan et al reported in their aforementioned study that their findings suggest that AI chatbots are an accurate and reliable supplementary source of medical information. However, they noted that the readability of these chatbots is limited and emphasized that they should not replace health professionals for personalized healthcare inquiries.[25]

In our study, the areas where these 3 chatbots lost points were primarily related to the absence of citations for source material, impacting both reliability and timeliness. In our study, we found that 100% of ChatGPT-3.5 responses and 99% of Google Bard responses lacked the criteria outlined by JAMA, such as authorship, references, sources, date indication, and ownership. In Perplexity, each response typically contained 5 or 6 sources, which were diverse in nature. These sources included videos, web pages from associations or hospitals intended for public education, scientific articles, as well as written texts. It can be assumed that the presence of videos and pictures will facilitate the reader's understanding. However, some of the sources were web pages of associations or hospitals designed to inform the public and could not provide data such as author and date. Across all 4 chatbots, the American Heart Association (AHA) CPR guideline was the most frequently cited source guide in response to queries, with the International Liaison Committee on Resuscitation (ILCOR) cited in 2 queries.

This study has several limitations. Firstly, the results cannot be generalized to all languages because CPR-related queries and responses were only assessed in English. Evaluating the accuracy and readability of responses to non-English queries would require investigation into a sufficiently large and diverse dataset. Secondly, there is currently no standardized methodology for evaluating chatbot responses. This lack of standardization leads to heterogeneity across studies and complicates comparisons between them. However, it is anticipated that a consensus on this matter will emerge in the future. Another limitation of our study is the potential inconsistency in responses from the chatbots. Several factors contribute to these inconsistencies. Some of them include; Contradictions in the data used to train the LLM. Variations in the wording used by different users to ask the same question. Differences in the context of the dialogue used to generate the chatbot response. Utilization of different decoding strategies and parameter values to introduce diversity in responses (such as Top-p, Top-k, temperature, etc).[43] The "top-p" value determines the number of possible words considered during text generation. When set to a low value, less likely words are excluded. Top-p is a method for regulating the randomness of generated text. Parameters like Temperature and Top-p can be adjusted to alter the language responses from chatbots. "Temperature" is a setting that influences randomness in word selection during text generation. Lower temperature values result in more predictable and consistent text, whereas higher values introduce more variability and creativity, albeit with potentially less consistency.[43–46] Therefore, it would be consider to repeat the current study using different "temperatures" in AI systems.[44] The date of output of responses from chatbots is indicated, but responses may change over time due to version changes and updates to chatbots. Evaluations only relate to the current versions and this should be taken into account. It crucial to recognize that, in their current state, chatbots should not serve as the sole authority in healthcare decision-making. They cannot replace consultations with medical professionals, especially concerning matters like evaluating treatment options and guiding patients in decision-making processes.[6,13,46]

5. Conclusion

When analyzing the responses to the query "What are the 100 most asked questions about cardiopulmonary resuscitation (CPR)?" from ChatGPT-3.5, Bard, Gemini, and Perplexity, it was found that the quality of the text content was at an intermediate level, and the readability was above the 6th-grade level. Considering that AI-based LLMs will increasingly serve as sources of information for patients, we believe that enhancing the quality, reliability, and readability of PEM will improve reader comprehension and the correct execution of CPR. Therefore, this could lead to an increased likelihood of survival for patients who receive bystander CPR. More comprehensive research is necessary to establish guidelines that ensure the reliability and accuracy of information generated by AI. Additionally, increasing users' awareness of the potential benefits and limitations of these tools is crucial.

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